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Quadratic magneto-optical Kerr effect in Co_2MnSi

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Quadratic magneto-optical Kerr effects (QMOKE) are investigated in epitaxial Co_2MnSi thin films as a function of the post-deposition annealing temperature. We determine the amplitude of the QMOKE signal for different annealing temperatures, which provide various degrees of $L2_1$ crystal ordering, as manifested by X-ray diffraction. We observe that QMOKE is significantly present only when the $L2_1$ ordering phase is also present, and that the QMOKE signal increases with the higher degree of $L2_1$ ordering. Additionally, we notice that the linear magneto-optical Kerr effect (LMOKE) decreases with higher annealing temperature. A comparison of the linear and quadratic MOKE contributions shows the amplitudes are of similar magnitude. From these results, we conclude that the presence of QMOKE is linked to the higher degree of $L2_1$ ordering in Co_2MnSi. © *2011 American Institute of Physics*. [doi:10.1063/1.3622512]

I. INTRODUCTION

Cobalt-based Heusler compounds have been intensively investigated because they are predicted to be half-metallic ferromagnets.[1] This property makes them promising candidates for applications in spintronic devices. Through the investigation of the magnetic, electronic, and structural properties of these materials, it has come to light that many Co-based Heusler compounds show strong second-order magneto-optical Kerr effects.[2] For example, this has recently been reported for Co_2FeSi,[3] Co_2MnSi,[4] Co_2MnGe,[5–7] and $Co_2FeSi_{0.5}Al_{0.5}$[8] epitaxial thin films.

While in (111)-oriented $Co_xMn_yGe_z$ the QMOKE was seen to strongly depend on the composition of the film,[6,7] all studies revealed that the magnetic properties of Heusler compounds strongly depend on the crystal structure. It is well known that Heusler compounds can exhibit various types of ordering, such as $B2$ (ordered Co sublattices, and statistical occupation of the two other crystallographic sites by Mn and Si in Co_2MnSi) and $L2_1$ (where the Mn and Si occupy distinct sublattices) ordering.[2] This ordering of the crystal structure can be improved with post-deposition annealing, an approach which has been demonstrated for several systems.[9–12] Gaier *et al.*[4] reported an improvement of $L2_1$ crystal ordering in Co_2MnSi with annealing temperature, and investigated the effect this had on the magnetic anisotropy, saturation magnetization, and exchange stiffness. The authors mentioned the presence of QMOKE but did not further investigate how it depended on the annealing temperature.

The present work quantitatively addresses the correlation between the second-order magneto-optical Kerr effect and the crystallographic structure of the compound. We investigate a series of Co_2MnSi thin-film samples with different post-deposition annealing temperatures, complementing our previous work.[4]

II. SAMPLE PREPARATION AND X-RAY DIFFRACTION

Co_2MnSi films were epitaxially grown by inductively coupled plasma-assisted magnetron sputtering with a (100) interface on a single-crystal MgO substrate covered with a 40 nm thick Cr buffer layer. The thickness of the Co_2MnSi films is 30 nm. To prevent oxidation, the thin-film stack is capped with a 1.3 nm thick Al protective layer. After deposition, the samples were annealed *in situ* at different temperatures, in order to achieve different degrees of ordering. The annealing temperatures ranged from 300 °C up to 500 °C.

All films were shown to be epitaxial single-crystal films from X-ray diffraction patterns. The $\theta - 2\theta$ scans show clearly the fundamental (400) reflection as well as the (200) superlattice reflection, indicating a $B2$ order (Fig. 1(a)). The data collected from a bare MgO substrate was subtracted from the 2θ scans. Due to this subtraction, the intensity of the peaks originating from the MgO substrate are greatly diminished. In order to assess the extent of $L2_1$ ordering with increasing annealing temperature, detailed X-ray diffraction measurements were performed. In Figs. 1(b) and 1(c) the pole figures (scan around the in-plane angle ϕ) of the fundamental (220) reflection and the (111) reflection are shown; the latter one is the characteristic superlattice diffraction peak of the $L2_1$ structure. For the lower annealing temperatures (300 °C and 350 °C), no (111) reflections are seen. The (111) peaks first appear for the sample annealed at 400 °C,

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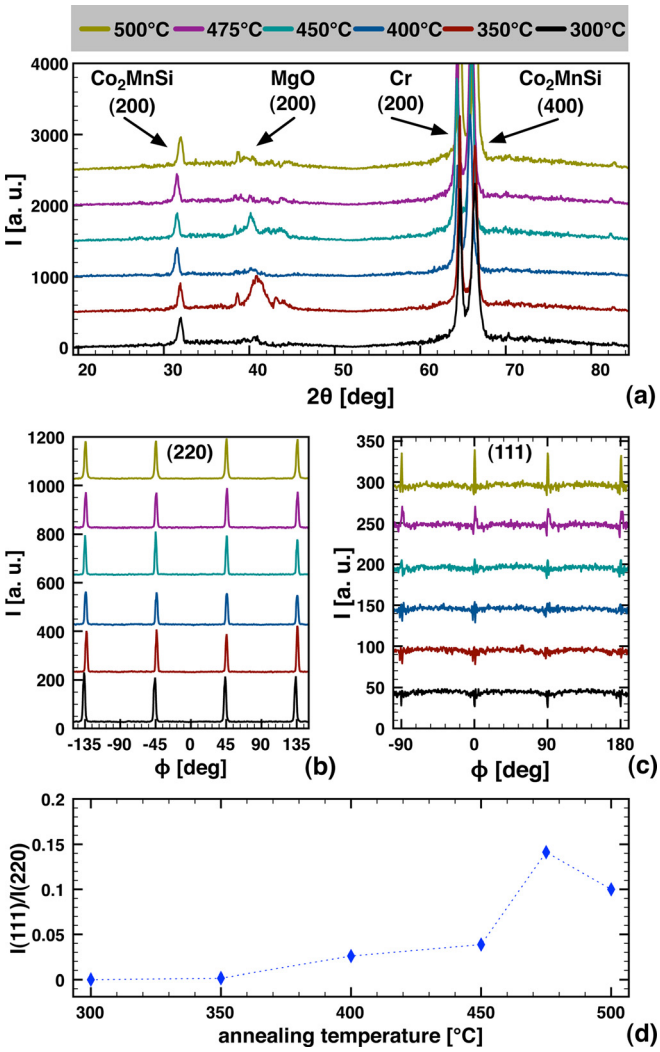

FIG. 1. (Color online) X-ray diffraction. (a) $\theta-2\theta$ scans for different post-deposition annealing temperatures. For clarity the intensity of a bare MgO substrate was subtracted. Peaks from the MgO substrate appear smaller than they are. (b), (c) Pole figures (scan around the in-plane angle ϕ) of the (220) and the (111) peak for different post-deposition annealing temperatures. (d) Ratio of the intensities of the (111) plane reflection to the (220) plane reflection as a function of the annealing temperature. The ratio $I_{(111)}/I_{(220)}$ is proportional to the degree of $L2_1$ long-range ordering.

and their intensity increases up to a maximum at an annealing temperature of 500 °C. A more quantitative specification is the ratio of the intensities of the (111) to the (220) plane reflections ($I_{(111)}/I_{(220)}$), which is proportional to the long range $L2_1$ order parameter.[4,13] Figure 1(d) presents this ratio as a function of annealing temperature, where an increase with rising annealing temperature is observed, with the highest ratio obtained for the sample annealed at 475 °C. This result is slightly different, but in general agreement to what was previously reported for Co_2MnSi.[4] Thus, we conclude that annealing at higher temperature promotes a higher degree of $L2_1$ ordering.

An analysis of the $I_{(200)}/I_{(220)}$ indicates the $B2$ degree of ordering (with respect to $A1$ disorder) does not significantly change over the range of annealing temperatures used here. As such, we infer that all trends observed below mainly stem from the modification of the $L2_1$ type of ordering.

III. QUADRATIC KERR EFFECTS

The Co_2MnSi thin films were investigated by means of magneto-optical Kerr effect (MOKE) magnetometry. The method used to characterize the QMOKE has been successfully used in several other works.[5,8,14,15] In this section, only a brief description of the underlying theory and the experimental method is given. More details on the instrumentation and experimental procedure can be found in our recent reports.[8,16]

Starting from the most general case, the longitudinal MOKE configuration (s-polarization and finite angle of incidence), the Kerr signal is determined by several contributions

$$\Theta_{Kerr} = \theta_L M_L + \theta_P M_P + \sum_{i,j} \theta_{ij} M_i M_j + \cdots, \qquad (1)$$

where θ_L, θ_P, and θ_{ij} are proportionality factors, containing the angle of incidence, index of refraction, and the magneto-optic constants. M_i and M_j are the magnetization components in the coordinate system of the incident light, where i,j can be L,T, and P. M_L is the in-plane magnetization component parallel to the plane of the incident light (longitudinal component), M_T is the in-plane magnetization component perpendicular to the plane of the incident light (transverse component), and M_P is the magnetization component out of the film plane (polar component). Since soft magnetic thin films are the subject of this investigation, strong demagnetizing fields coerce the magnetization to lie into the film plane,[17] which allows us to neglect the out-of-plane component M_P.

In such a case, the measured Kerr signal is determined by the linear longitudinal term (LMOKE) and the second-order terms (QMOKE), the summation of Eq. (1). The exact Kerr angle can be calculated by expanding the dielectric tensor as a function of the magnetization up to the second order using a wave equation formalism describing light inside a multilayer structure (for a detailed description, see Refs. 14, 18, and 19). For the investigated system with a cubic symmetry, a (100) interface plane and s-polarized light, the complex Kerr angle is given by[14]

$$\begin{aligned}\Theta_{Kerr} = {} & BKM_L + A\left(-\frac{K^2}{\epsilon} + 2G_{44} + \frac{\Delta G}{2}\right)M_L M_T \\ & - A\frac{\Delta G}{2}\cos(4\alpha)M_L M_T \\ & - A\frac{\Delta G}{4}\sin(4\alpha)(M_L^2 - M_T^2),\end{aligned} \qquad (2)$$

where ϵ is the dielectric constant, K the linear magneto-optical constant, and $\Delta G = G_{11} - G_{12} - 2G_{44}$, with G_{11}, G_{12}, and G_{44} being the remaining elements of the quadratic magneto-optical tensor. The angle α is the orientation of the crystallographic axes with respect to the plane of incidence (see Fig. 3(a)). The factors A and B are pure optical factors that are even and odd functions of the angle of incidence. At perpendicular incidence, the LMOKE contribution vanishes, but for an in-plane magnetized film there is still a detectable signal from the second-order contributions. In Eq. (2) three different

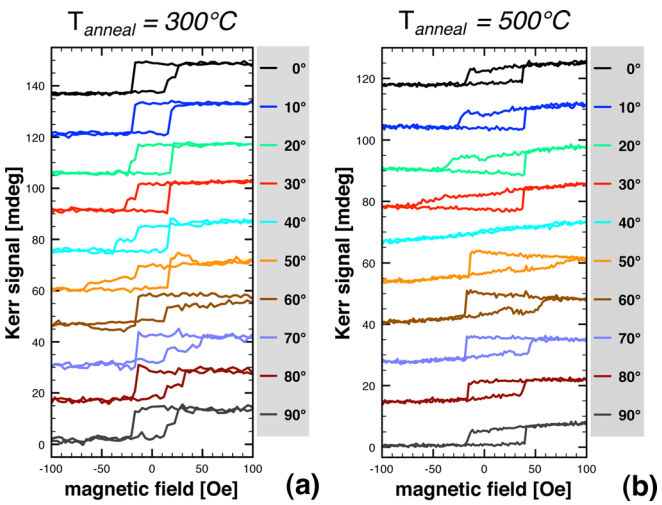

FIG. 2. (Color online) Magnetization reversal curves for (a) a sample annealed at 300 °C and (b) annealed at 500 °C for different sample orientations with respect to the magnetic field, ranging from 0° to 90°.

contributions to the total Kerr signal appear: (1) the linear contribution proportional to M_L (LMOKE), (2) the contribution proportional to the product of $M_L M_T$, and (3) the contribution proportional to the difference of the squares of the longitudinal and transverse magnetization components $M_L^2 - M_T^2$. It is obvious that both second-order contributions depend on the sample orientation and on the value of ΔG. The $M_L M_T$ contribution also has a constant offset, which depends on all magneto-optical constants, but not on α (isotropic QMOKE).

To separate the different contributions, we used a Kerr-effect magnetometer described elsewhere.[16] The setup uses probing light of a wavelength of 638 nm. All measurements are limited to the Kerr rotation and carried out at room temperature. Saturation fields are applied in the eight directions indicated in Fig. 3(a). In these directions either the $M_L M_T$ contribution or the $M_L^2 - M_T^2$ contribution is zero and the remaining contribution can be extracted.[8] Rotating the sample in-plane, we can obtain the anisotropy of the QMOKE contributions. This procedure was applied to the Co_2MnSi thin films of interest here.

IV. RESULTS

A. Magnetization reversal

Magnetization reversal loops were obtained as a function of sample orientation with respect to the applied magnetic field at an angle of incidence of 45°. Figure 2 shows MOKE magnetization reversal curves for different orientations for samples annealed at 300 °C (a) and 500 °C (b), respectively. The Kerr signal is rather small compared to the noise level for all the samples. The shape of the magnetization reversal loops changes with the sample orientation mainly due to the magnetic anisotropy. For the sample annealed at 300 °C all loops are of a similar square shape, indicating a uniform anisotropy distribution. For the sample annealed at 500 °C the magnetic easy axis is oriented along 0° and 90° (corresponding to ⟨110⟩ Co_2MnSi directions), whereas the hard axis is found at an orientation of 45° (corresponding to ⟨100⟩ Co_2MnSi directions). This behavior is expected for a system with cubic symmetry. The shape of the reversal loops is influenced by the magnetic anisotropy as well as the nucleation of magnetic domains and the movement of magnetic domain walls. For both samples, one can observe an additional asymmetry in the reversal loops, which is due to the QMOKE. However, the complex interplay of anisotropy and QMOKE makes it nearly impossible to draw precise conclusions about the magnetization state of the sample from the measured Kerr signal. Only a symmetrization/antisymmetrization procedure allows one to extract the LMOKE and QMOKE contributions from the measured signal.[3] However, this procedure provides limited insight, as it does not allow one to identify the different QMOKE contributions presented in Eq. (2). The small signal-to-noise ratio is an additional source of error in the symmetrization procedure. As such, we instead rely on QMOKE measurements in saturation, described below.

B. Quadratic MOKE in saturation

Measurements in saturation along the eight fields shown in Fig. 3(a) were used to separate the linear and quadratic MOKE contributions. As the measurements are done in saturation, they are insensitive to effects such as the formation and movement of domains, or the magnetic anisotropy as visible in the reversal loops. Figure 3(b) presents a typical

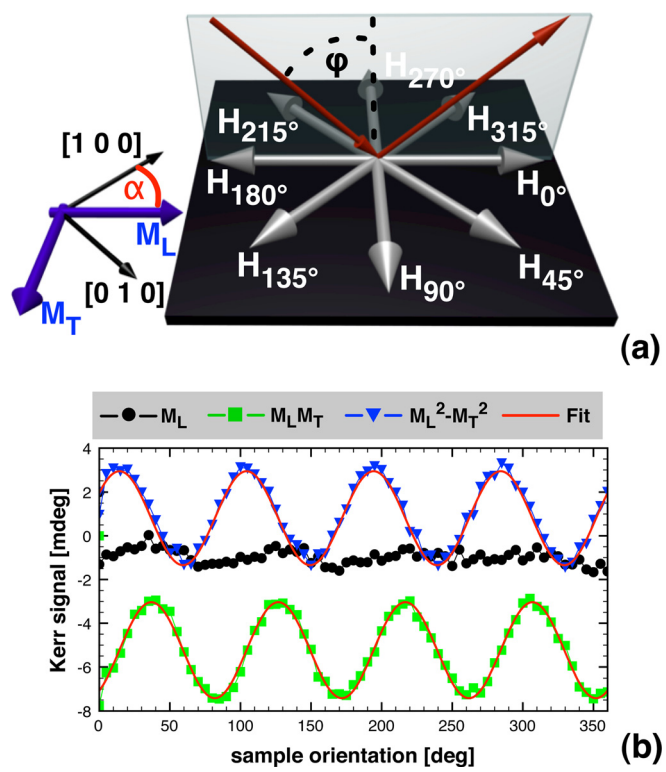

FIG. 3. (Color online) (a) Schematic drawing of the QMOKE measurements in saturation. Saturation fields are applied in the eight in-plane directions to separate the different quadratic contributions. (b) The separated QMOKE contributions as a function of the sample orientation α for a sample annealed at 475 °C recorded at perpendicular incidence ($\varphi = 0°$). M_L contribution (black dots), $M_L M_T$ contribution (green squares), $M_L^2 - M_T^2$ contribution (blue triangles), fit function according to Eq. (2) (red lines).

result for the sample annealed at 475 °C. The signal was recorded at carefully aligned perpendicular incidence, where the linear longitudinal contribution vanishes. Both quadratic contributions follow a sin (cos) function as expected from Eq. (2). Fitting the experimental data, the amplitudes of the sin (cos) and the constant offset were extracted. These values are directly correlated to the second-order magneto-optical constants, especially the ΔG value and thus the strength of the QMOKE.

Figure 4 shows these amplitudes and offset as a function of the annealing temperature. The QMOKE measurements for the low annealing temperatures (300 °C and 350 °C) show no clear sinusoidal oscillations of the signal. The QMOKE amplitudes obtained from fitting the data are rather small, such that we assume them to be zero within the experimental error. For the higher annealing temperatures ($\geq$400 °C) the values for the amplitude of the $M_L M_T$ and the $M_L^2 - M_T^2$ contributions are the same within the experimental error for each annealing temperature. This is expected from the model (also see Ref. 3), and this is a good indication that the cubic model we applied is valid. The values of the amplitudes increase up to the maximum value of 2.1 mdeg for an annealing temperature of 475 °C. This value is comparable to the QMOKE amplitude previously reported in a Co_2MnSi thin film.[16] As shown in the X-ray data of Fig. 1, the $L2_1$ structure is not present for annealing temperatures below 400 °C, and increases for the higher annealing temperatures. We can summarize that for samples without $L2_1$ structure no or only very small anisotropic QMOKE is present and emerges with higher degrees of $L2_1$ ordering. In particular, the ΔG value increases with the higher degree of ordering.

The offset of the $M_L M_T$ contribution (red dots in Fig. 4) is larger than the amplitudes, and there is already a reasonable offset visible for the lower annealing temperatures ($\leq$400 °C), which leads to the QMOKE visible in the magnetic reversal loops (see Fig. 2(a)). Generally, the offset follows the same trend as the amplitudes. According to Eq. (2) the linear magneto-optical constant K and G_{44} contribute to the offset. Since K is in general a small number ($K \ll 1$) and of the order of the quadratic magneto-optical constants (see below), $\frac{K^2}{\epsilon}$ can be neglected in a first approximation. The difference between amplitudes and offset is now only given by the G_{44} factor. The difference between offset and amplitude is not changing drastically, so we can assume G_{44} as not changing drastically with annealing temperature. We conclude the ΔG magneto-optical anisotropy factor is increasing with annealing temperature, and predominantly accounts for the trends in the overall QMOKE with annealing temperatures. This increased magneto-optical anisotropy is probably rooted in the higher degree of atomic ordering, and modified point symmetry, of the $L2_1$ superstructure.

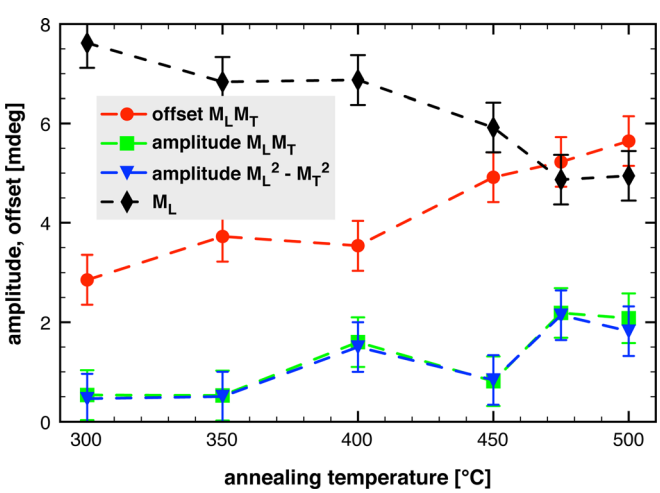

FIG. 4. (Color online) The M_L contribution (black diamonds) at angle of incidence $\varphi = 45°$, the amplitudes and offset of the $M_L M_T$ contribution (green squares and red dots) and the amplitude of the $(M_L^2 - M_T^2)$ (blue triangles) at perpendicular incidence $\varphi = 0°$ as a function of the post-deposition annealing temperature.

At this point it is worth noting that the linear magneto-optical signal, obtained at an angle of incidence of $\varphi = 45°$, decreases with the annealing temperature (black diamonds in Fig. 4), which is quite unexpected as the second-order effect increases.

Finally, we want to lay out an open question for future investigation. In Fig. 3(b) the M_L contribution is approximately zero, as these curves were recorded at perpendicular incidence. The observed deviation from zero is about 1 mdeg. Perpendicular incidence can be realized within $\Delta\varphi \approx 1°$, which would only lead to a Kerr rotation of approximately 0.15 mdeg, estimated from the value measured at 45° incidence. Furthermore this curve also reveals a small-amplitude periodic behavior with an amplitude of about 0.3 mdeg, even though the linear MOKE is not expected to show anisotropy in a cubic material (see Eq. (2)). This effect was previously observed (though not pointed out) for Co_2 $FeAl_{0.5}$ $Si_{0.5}$ (Ref. 8) and Co_2MnSi (Ref. 16) films.

A possible reason may be a slight tilt of the sample plane, which would lead to a variation in the angle of incidence during sample rotation. But, we have verified that the sample plane can be aligned with an accuracy better than 1°. The observed deviations are too large to be simply explained by misalignment, and the origin of this periodicity is yet unclear and requires further careful exploration. Moreover, the linear MOKE and the quadratic MOKE are of the same order of magnitude. The question arises whether a non-negligible third order effect may be present. At this point a more rigorous study with advanced theoretical support must be performed to shed light on this issue.

V. CONCLUSIONS

In summary, we investigated the dependence of the quadratic magneto-optical Kerr effect (QMOKE) on the post-deposition annealing temperature of Co_2MnSi films. X-ray diffraction reveals that the $L2_1$ ordered structure is present in samples annealed at a temperature higher than 350 °C. We observe that only isotropic QMOKE is present for the lower annealing temperatures. The anisotropic contribution appears for the sample annealed at 400 °C, and increases for higher annealing temperatures. $L2_1$ ordering as well as the QMOKE amplitudes show a maximum for the sample annealed at 475 °C. We also find that for samples exhibiting QMOKE the measurements in saturation can be fit very well by the theoretical prediction for a cubic system. The linear magneto-optical constant was found to be in general quite small in this system, and slightly decreases with higher annealing temperature.

Comparing the QMOKE amplitude with the LMOKE amplitude, we can say that the QMOKE is of a similar magnitude in highly ordered Co_2MnSi. Based on these results and the quasi-constant degree of $B2$ ordering, we link the higher degree of $L2_1$ ordering with the presence of QMOKE, even though the linear MOKE is quite small. In particular, we observe the cubic magneto-optical term ΔG increases with $L2_1$ order. A similar result of increasing QMOKE with higher ordering was recently observed for $Co_2FeSi_{0.5}Al_{0.5}$ thin films.[8] However, in those $Co_2FeAl_{0.5}Si_{0.5}$ films, both the iso- and anisotropic QMOKE grew concurrently. Here we observe only the presence of isotropic QMOKE in some of our Co_2MnSi thin films and the anisotropic QMOKE emerges for higher annealing temperature.

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